

江苏省 2014 年普通高校专转本统一考试

高等数学 答案解析

一、选择题

1. 解: 因为 $x=1$ 是函数 $f(x) = \frac{x^2 - 4x + a}{x^2 - 3x + 2}$ 的可去间断点

$$\text{所以 } \lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{x^2 - 4x + a}{x^2 - 3x + 2} \text{ 存在}$$

$$\text{因为 } \lim_{x \rightarrow 1} (x^2 - 3x + 2) = 0, \text{ 所以必有 } \lim_{x \rightarrow 1} (x^2 - 4x + a) = 0$$

$$\text{即 } 1 - 4 + a = 0, \quad a = 3$$

2. 解: $y = x^4 - 2x^3$ 的定义域为 $(-\infty, +\infty)$

$$y' = 4x^3 - 6x^2, \quad y'' = 12x^2 - 12x = 12x(x-1)$$

因为在 $[0, 1]$ 上 $y'' < 0$, 所以在 $[0, 1]$ 曲线是凸的

3. 解: $f(x) = (x \sin x)' = \sin x + x \cos x$

$$\int f''(x) = f'(x) + c = (\sin x + x \cos x)' + c = 2 \cos x - x \sin x + c$$

4. 解: 方程两边对 x 求偏导数得 $3z^2 \frac{\partial z}{\partial x} - 3yz - 3xy \frac{\partial z}{\partial x} + 3x^2 = 0$

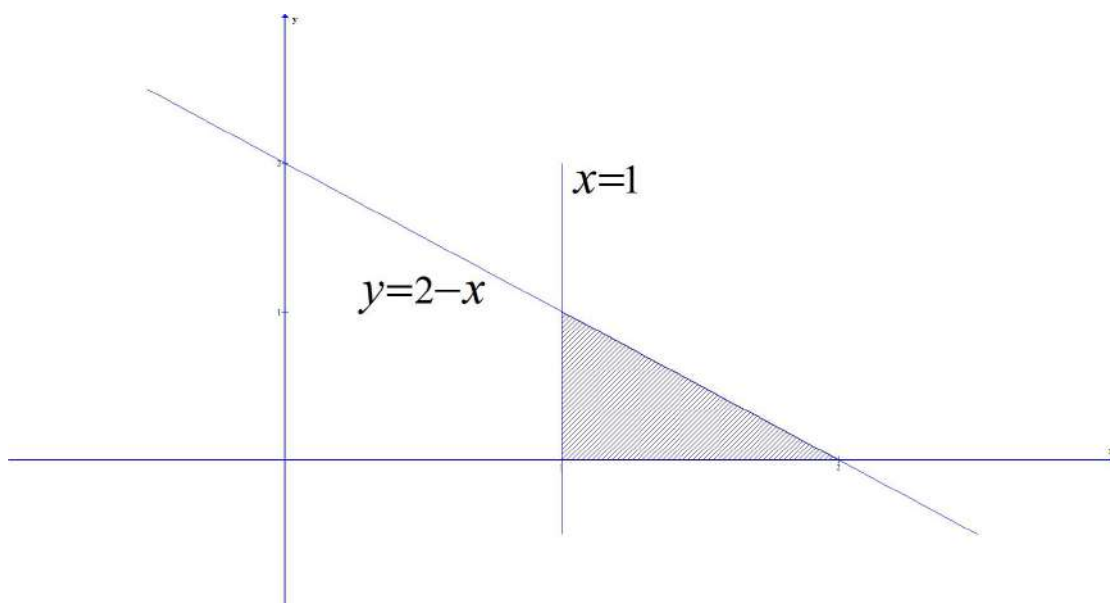
$$\text{所以 } \frac{\partial z}{\partial x} = \frac{yz - x^2}{z^2 - xy}, \quad \left. \frac{\partial z}{\partial x} \right|_{\substack{x=1 \\ y=0}} = -\frac{1}{z^2}$$

将 $x=0, y=1$ 代入原方程得 $z=1$

$$\text{所以 } \left. \frac{\partial z}{\partial x} \right|_{\substack{x=1 \\ y=0}} = -\frac{1}{z^2} \Big|_{z=1} = -1$$

5. 解: 积分区域 $D: \begin{cases} 0 \leq y \leq 2-x \\ 1 \leq x \leq 2 \end{cases}$

$$\text{交换积分次序得 } \int_1^2 dx \int_0^{2-x} f(x, y) dy = \int_0^1 dy \int_1^{2-y} f(x, y) dx$$



6. 解：对于 A, $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ 收敛

对于 B, $\left| \frac{\sin n}{n^2} \right| \leq \frac{1}{n^2}$, 而 $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 收敛, 所以 $\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$ 收敛

对于 C, 因为 $\sum_{n=1}^{\infty} \frac{1}{2^n}$, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 均收敛, 所以 $\sum_{n=1}^{\infty} \left(\frac{1}{2^n} + \frac{1}{n^2} \right)$

应选 D

二、填空题

7. 解：因为 $\lim_{x \rightarrow \infty} \left(1 - \frac{2}{x} - 1 \right) \cdot x = -2$

所以 $\lim_{x \rightarrow \infty} \left(1 - \frac{2}{x} \right)^x = e^{-2}$, 所以曲线的水平渐近线为 $y = e^{-2}$

8. 解： $f'(x) = 3ax^2 - 18x + 12$

因为 $f(x)$ 在 $x = 2$ 处取得极小值, 所以应有 $f'(2) = 0$

即 $12a - 36 + 12 = 0$, 所以 $a = 2$

于是 $f'(x) = 6x^2 - 18x + 12 = 6(x-1)(x-2)$

令 $f'(x)=0$ 得 $x=1, x=2$, 由题意可知 $f(x)$ 在 $x=1$ 处取得极大值

$$\begin{aligned} 9. \text{ 解: } \int_{-1}^1 (x^3+1)\sqrt{1-x^2} dx &= \int_{-1}^1 x^3 \sqrt{1-x^2} dx + \int_{-1}^1 \sqrt{1-x^2} dx \\ &= 0 + 2 \int_0^1 \sqrt{1-x^2} dx = 2 \cdot \frac{1}{4} \pi = \frac{\pi}{2} \end{aligned}$$

$$10. \text{ 解: } \frac{\partial z}{\partial x} = \frac{1}{1+\frac{y^2}{x^2}} \cdot \frac{-y}{x^2} = \frac{-y}{x^2+y^2}, \quad \frac{\partial z}{\partial y} = \frac{1}{1+\frac{y^2}{x^2}} \cdot \frac{1}{x} = \frac{x}{x^2+y^2}$$

$$\text{所以 } dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = \frac{-y}{x^2+y^2} dx + \frac{x}{x^2+y^2} dy$$

11. 略

$$12. \text{ 解: 令 } \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}(x)}{u_n(x)} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x-1)^{n+1}}{\sqrt{n+1}} \cdot \frac{\sqrt{n}}{(x-1)^n} \right| = |x-1| < 1, \text{ 得 } 0 < x < 2$$

$$\text{当 } x=0 \text{ 时, 级数为 } \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}} \text{ 收敛}$$

$$\text{当 } x=2 \text{ 时, 级数为 } \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \text{ 发散}$$

所以收敛域为 $[0, 2)$

三、计算题

$$\begin{aligned} 13. \text{ 解: } \lim_{x \rightarrow 0} \left(\frac{1}{x \arcsin x} - \frac{1}{x^2} \right) &= \lim_{x \rightarrow 0} \frac{x - \arcsin x}{x^2 \arcsin x} = \lim_{x \rightarrow 0} \frac{x - \arcsin x}{x^3} \\ &= \lim_{x \rightarrow 0} \frac{1 - \frac{1}{\sqrt{1-x^2}}}{3x^2} = \lim_{x \rightarrow 0} \frac{\sqrt{1-x^2} - 1}{3x^2 \sqrt{1-x^2}} \\ &= \lim_{x \rightarrow 0} \frac{-\frac{1}{2}x^2}{3x^2} = -\frac{1}{6} \end{aligned}$$

$$14. \text{ 解: } \frac{dx}{dt} = e^{2t} + 2(t+1)e^{2t} = (2t+3)e^{2t}$$

$$e^y + ty = e \text{ 两边对 } t \text{ 求导得 } e^y \frac{dy}{dt} + y + t \frac{dy}{dt} = 0$$

$$\text{所以 } \frac{dy}{dt} = \frac{-y}{e^y + t}$$

$$\text{所以 } \frac{dy}{dx} = \frac{-y}{e^{2t}(e^y + t)(2t + 3)}$$

当 $t = 0$ 时, 由 $e^y + ty = e$ 可知 $y = 1$

$$\text{于是 } \left. \frac{dy}{dx} \right|_{t=0} = \left. \frac{-y}{3e^y} \right|_{y=1} = \frac{-1}{3e}$$

$$\begin{aligned} 15. \text{ 解: } \int x \ln^2 x dx &= \frac{1}{2} \int \ln^2 x dx^2 = \frac{1}{2} x^2 \ln^2 x - \frac{1}{2} \int x^2 d \ln^2 x = \frac{1}{2} x \ln^2 x - \int x \ln x dx \\ &= \frac{1}{2} x^2 \ln^2 x - \frac{1}{2} \int \ln x dx^2 = \frac{1}{2} x^2 \ln^2 x - \frac{1}{2} x^2 \ln x + \frac{1}{2} \int x^2 d \ln x \\ &= \frac{1}{2} x^2 \ln^2 x - \frac{1}{2} x^2 \ln x + \frac{1}{2} \int x dx \\ &= \frac{1}{2} x^2 \ln^2 x - \frac{1}{2} x^2 \ln x + \frac{1}{4} x^2 + c \end{aligned}$$

$$16. \text{ 解: 令 } \sqrt{2x-1} = t, \text{ 则 } x = \frac{t^2+1}{2}, \text{ 当 } x = \frac{1}{2} \text{ 时 } t = 0, \text{ 当 } x = \frac{5}{2} \text{ 时 } t = 2$$

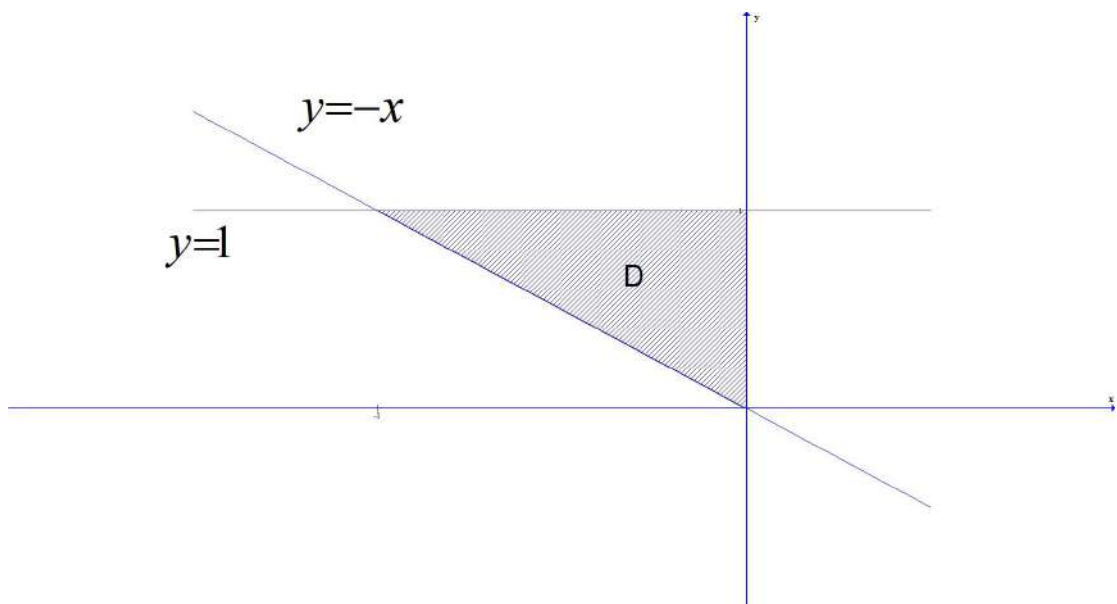
$$\begin{aligned} \int_{\frac{1}{2}}^{\frac{5}{2}} \frac{\sqrt{2x-1}}{2x+3} dx &= \int_0^2 \frac{t}{t^2+4} d \frac{t^2+1}{2} = \int_0^2 \frac{t^2+4-4}{t^2+4} dt \\ &= \int_0^2 dt - 4 \int_0^2 \frac{1}{t^2+4} dt = t \Big|_0^2 - 2 \arctan \frac{t}{2} \Big|_0^2 = 2 - \frac{\pi}{2} \end{aligned}$$

17. 略

$$18. \text{ 解: } \frac{\partial z}{\partial x} = \cos x f_1' + 2x f_2'$$

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} &= \cos x [f_{11}'' \cdot 0 + f_{12}'' \cdot (-2y)] + 2x [f_{21}'' \cdot 0 + f_{22}'' \cdot (-2y)] \\ &= -2y \cos x f_{12}'' + 4xy f_{22}'' \end{aligned}$$

$$\begin{aligned} 19. \text{ 解: } \iint_D (x+y) dx dy &= \int_0^1 dy \int_{-y}^0 (x+y) dx = \int_0^1 dy \cdot \left(\frac{1}{2} x^2 + yx \right) \Big|_{-y}^0 \\ &= \frac{1}{2} \int_0^1 y^2 dy = \frac{1}{6} y^3 \Big|_0^1 = \frac{1}{6} \end{aligned}$$



20. 解：对应齐次方程 $y'' - 2y' = 0$ 的特征方程为 $r^2 - 2r = 0$ ，特征根 $r_1 = 0, r_2 = 2$

$$\text{通解 } \bar{y} = c_1 + c_2 e^{2x}$$

$$\text{设原方程的一个特解为 } y^* = x(ax + b)e^{2x} = (ax^2 + bx)e^{2x}$$

$$y^{*'} = (2ax + b)e^{2x} + 2(ax^2 + bx)e^{2x}$$

$$y^{*''} = 2ae^{2x} + 2(2ax + b)e^{2x} + 2(2ax + b)e^{2x} + 4(ax^2 + bx)e^{2x}$$

将 y^* 代入原方程得

$$2ae^{2x} + 4(2ax + b)e^{2x} + 4(ax^2 + bx)e^{2x} - 2(2ax + b)e^{2x} - 4(ax^2 + bx)e^{2x} = xe^{2x}$$

$$\text{即 } (4ax + 2a + 2b) = x$$

$$\text{比较两边同类项系数得 } a = \frac{1}{4}, b = -\frac{1}{4}$$

$$\text{所以 } y^* = \left(\frac{1}{4}x^2 - \frac{1}{4}x \right) e^{2x}$$

$$\text{原方程通解 } y = \bar{y} + y^* = c_1 + c_2 e^{2x} + \left(\frac{1}{4}x^2 - \frac{1}{4}x \right) e^{2x}$$

四、证明题

21. 证明：令 $F(x) = x \ln x - 3$ ，则 $F(x)$ 在 $[2, 3]$ 上连续

$$F(2) = 2\ln 2 - 3 < 0, \quad F(3) = 3\ln 3 - 3 > 0$$

由零点定理可知, 方程在 $(2, 3)$ 内至少有一根

$F'(x) = \ln x + 1$, 当 $x \in (2, 3)$ 时, $F'(x) > 0$, 所以在 $(2, 3)$ 内 $F(x)$ 单调递增

方程在 $(2, 3)$ 内最多只有一根

综上所述, 方程 $x \ln x = 3$ 在区间 $(2, 3)$ 内有且仅有一个实根

22. 证明: 令 $F(x) = e^x - 1 - \frac{1}{2}x^2 - \ln(x+1)$, 则 $F(0) = 0$

$$F'(x) = e^x - x - \frac{1}{x+1}, \quad \text{则 } F'(0) = 0$$

$$F''(x) = e^x - 1 + \frac{1}{(x+1)^2}$$

当 $x > 0$ 时 $F''(x) > 0$, 所以当 $x > 0$ 时 $F'(x)$ 单调递增, $F'(x) > F'(0) = 0$

由 $F'(x) > 0$ 又可知 $F(x)$ 单调递增, 所以 $F(x) > F(0) = 0$

即 $e^x - 1 - \frac{1}{2}x^2 - \ln(x+1) > 0$, 所以 $e^x - 1 > \frac{1}{2}x^2 + \ln(x+1)$

五、综合题

23. 解: (1) $y' = -2x$, $k_{\text{切}} = y'|_{x=1} = -2$, 切线方程为 $y = -2(x-1)$

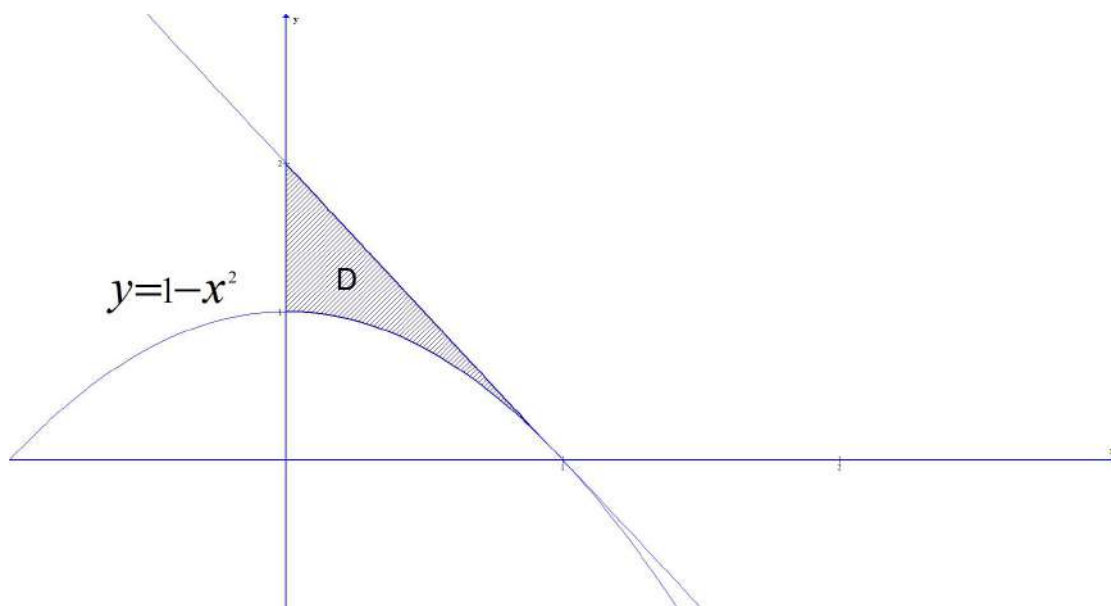
$$S = \iint_D dx dy = \int_0^1 dx \int_{1-x^2}^{-2(x-1)} dy = \int_0^1 (-2x + 1 + x^2) dx$$

$$= \left(-x^2 + x + \frac{1}{3}x^3 \right) \Big|_0^1 = \frac{1}{3}$$

$$(2) V = \pi \int_0^2 \left(\frac{2-y}{2} \right)^2 dy - \pi \int_0^1 (\sqrt{1-y})^2 dy$$

$$= \frac{1}{4} \pi \int_0^2 (y-2)^2 d(y-2) + \pi \int_0^1 (y-1) d(y-1)$$

$$= \frac{1}{12} \pi (y-2)^3 \Big|_0^2 + \frac{1}{2} \pi (y-1)^2 \Big|_0^1 = \frac{1}{6} \pi$$



24. 解: (1) 方程 $\int_0^x t\varphi(t)dt = 1 - \varphi(x)$ 两边对 x 求导得 $x\varphi(x) = -\varphi'(x)$

$$\text{即 } \varphi'(x) + x\varphi(x) = 0$$

$$\text{通解 } \varphi(x) = e^{-\int x dx} \left[\int 0 \cdot e^{\int x dx} dx + c \right] = ce^{-\frac{1}{2}x^2}$$

由原方程可知 $\varphi(0) = 1$, 代入通解得 $c = 1$, 所以 $\varphi(x) = e^{-\frac{1}{2}x^2}$

$$(2) f(x) = \begin{cases} \frac{e^{-\frac{1}{2}x^2} - 1}{x^2} & x \neq 0 \\ -\frac{1}{2} & x = 0 \end{cases}$$

$$\text{因为 } \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{e^{-\frac{1}{2}x^2} - 1}{x^2} = \lim_{x \rightarrow 0} \frac{-\frac{1}{2}x^2}{x^2} = -\frac{1}{2} = f(0)$$

所以 $f(x)$ 在 $x = 0$ 处连续

$$\begin{aligned} \text{又因为 } f'(0) &= \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{\frac{e^{-\frac{1}{2}x^2} - 1}{x^2} + \frac{1}{2}}{x} \\ &= \lim_{x \rightarrow 0} \frac{2e^{-\frac{1}{2}x^2} - 2 + x^2}{2x^3} = \lim_{x \rightarrow 0} \frac{-2xe^{-\frac{1}{2}x^2} + 2x}{6x^2} \\ &= -\frac{1}{3} \lim_{x \rightarrow 0} \frac{e^{-\frac{1}{2}x^2} - 1}{x} = -\frac{1}{3} \lim_{x \rightarrow 0} \frac{-\frac{1}{2}x^2}{x} = 0 \text{ 存在} \end{aligned}$$

所以 $f(x)$ 在 $x=0$ 处可导



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